

Improved Delayed Neutron Group Parameters for Molten Salt Reactors

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Delayed neutron precursors (DNPs) emit delayed neutrons

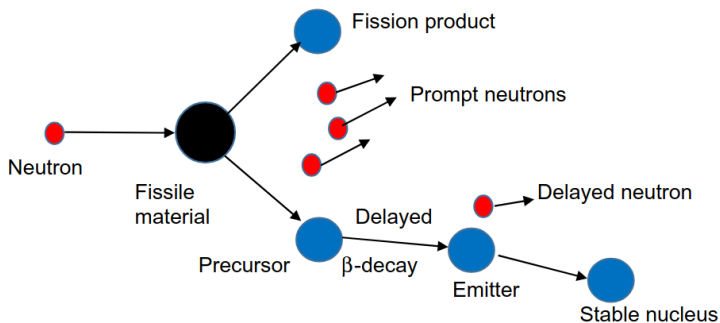


Figure 1: The process of delayed neutron emission from [17].

Delayed neutron precursors (DNPs) emit delayed neutrons

Delayed neutron precursors (DNPs) are nuclides that undergo β^- decay, producing a daughter, called an emitter, which emits a delayed neutron [21].

Relevant data for DNPs includes:

$P_{n,i}$ = the emission probability [#]

τ_i = the half-life of DNP i [s]

$\chi_{d,i}$ = the delayed neutron energy spectrum from DNP i [-]

N_i = the number of atoms of i [#].

Accurately modeling the DNPs is important for transient simulations, which are important for safety, security, and safeguards.

DNP groups approximate DNPs

A small number of artificial DNPs, known as DNP groups, are used to approximate the DNPs to reduce computational cost.

The three parameters for DNP groups are:

$\bar{\nu}_{d,k}$ = the delayed neutron yield of DNP group k [-]

τ_k = the half-life of DNP group k [s]

$\chi_{d,k}$ = the delayed neutron energy spectrum of DNP group k [-]

DNP groups are commonly used to approximate the DNPs [3].

Moltres provides multiphysics simulation capabilities

Moltres is a multiphysics reactor simulation tool built on the MOOSE framework [18].

Moltres enables:

- Multigroup neutron diffusion
- DNP recirculation
- Heat transfer and fluid flow

Moltres will be used for the proposed safety, security, and safeguards transient simulations.

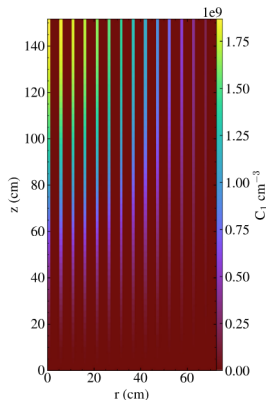


Figure 2: The longest-lived DNP group concentration in the 2D axisymmetric Moltres MSRE model from [18].

MSR phenomena are not captured in conventional parameters

Fluid-fueled molten salt reactor (MSR) phenomena that affect DNP group parameters:

- Recirculation of fuel salt
- Elemental chemical removal via reprocessing

Conventional DNP group parameters do not account for MSR recirculation or chemical removal.

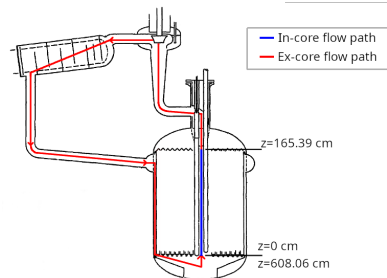


Figure 3: The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [10].



This work will capture MSR phenomena in the DNP groups

The primary objective of this work is to generate improved DNP group parameters for MSRs.

The objectives of this work are to:

-  Create a Python package (Molten Salt Delayed Neutron (MoSDeN)) for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems



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Two approaches can calculate the delayed neutron count rate

DNP group parameter calculation follows the following steps:

- ① Irradiate a fissile sample
- ② Measure/calculate a delayed neutron count rate
- ③ Determine the DNP group parameters that best fit that count rate

The specific steps depend on which approach is used.

Macroscopic approach

The delayed neutron count rate is *measured*.

Microscopic approach

The delayed neutron count rate is *calculated*.

The macroscopic approach irradiates a sample physically

The delayed neutron count rate ($\dot{n}_d$) is directly measured after sample irradiation.

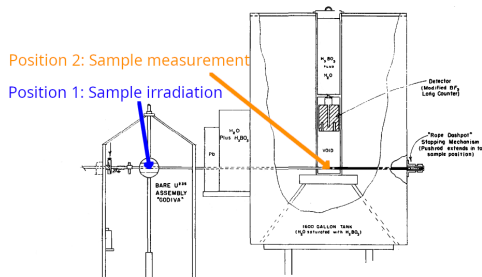


Figure 4: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [16].



The microscopic approach irradiates a sample computationally

The delayed neutron count rate is calculated from aggregated experimental data.

$$\dot{n}_d(t) = \sum_i^I P_{n,i} \lambda_i N_i(t), \quad (1)$$

where

I = the number of DNPs [#]

$\dot{n}_d$ = the delayed neutron count rate [$\# \cdot s^{-1}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from decay of nuclide i , or the emission probability [#]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of atoms of DNP i [#].



There is a general equation for sample irradiation

The I DNPs are approximated by K groups.

Intermediate Irradiation

$$\dot{n}_d(T, t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}) \quad (2)$$

$\dot{F}_s$ = The fission rate in the sample [$\# \cdot s^{-1}$]

F_s = The number of fissions in the sample [$\#$]

T = The time over which the sample is irradiated [s]



There are two common irradiation types used to generate groups

Intermediate Irradiation

$$\dot{n}_d(T, t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \left(1 - e^{-\lambda_k T}\right) \quad (3)$$

Pulse irradiation ($T \rightarrow 0$)

$$\dot{n}_d(t) \approx F_s \sum_{k=1}^K \bar{\nu}_{d,k} \lambda_k e^{-\lambda_k t} \quad (4)$$

Saturation irradiation ($T \rightarrow \infty$)

$$\dot{n}_d(t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \quad (5)$$

The DNP parameters $\bar{\nu}_{d,k}$ and λ_k make $\dot{n}_d(t)$ match using exponential peeling [5, 15], least squares [4, 16], or half-lives of specific DNPs [27].

Fuel salt recirculation affects concentration of DNPs

Concentrations of nuclides vary by up to 67% with and without accounting for fuel movement [30].

In-core and ex-core salt recirculation can be modeled using:

- **Scaled flux method** [2]
- **Piecewise 0D method**
- Spatially resolved method [30, 28]

Conventional parameters do not account for fuel recirculation effects.

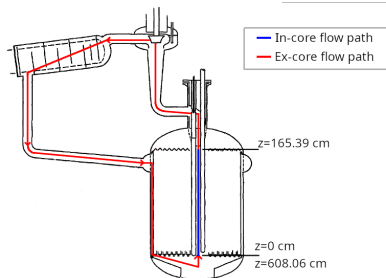


Figure 5: The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [10].



Chemical reprocessing affects DNPs concentrations

Online chemical reprocessing includes:

- Fresh fuel addition
- Fission product removal

Conventional DNP group parameters do not account for these effects.

For example, a reprocessing scheme with 100% of all DNPs chemically removed should not use conventional DNP groups.



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MoSDeN generates improved DNP group parameters

MoSDeN is an open-source Python package in development that will enable users to generate improved DNP group parameters.

MoSDeN currently enables:

- Use of various data inputs for half-lives (τ), emission probabilities (P_n), and cumulative fission yields (CFY)
 - ENDF-6 formatted data
 - OpenMC chain data
 - Custom data
- The microscopic approach using the 0D scaled model
 - Equilibrium approximated
 - Scaled flux method
 - Non-linear least squares
 - Generic fissile nuclide
 - Generic neutron irradiation energy

MoSDeN currently offers the 0D scaled model, which only enables saturation irradiations to be modeled.



The saturation irradiation is a non-linear least squares problem

The group parameters are calculated by equating the delayed neutron count rates.

$$\dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \approx \sum_i^I P_{n,i} \lambda_i N_i(t) \quad (6)$$

$$\dot{F}_s \begin{bmatrix} e^{-\lambda_1 t_0} & \dots & e^{-\lambda_K t_0} \\ \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix} \begin{bmatrix} \bar{\nu}_{d,1} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix} \approx \begin{bmatrix} \dot{n}_d(t_0) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix} \quad (7)$$

$$A\vec{x} \approx \vec{b} \quad (8)$$

The residual best fits the delayed neutron count rate

The relative residual provides a better fit for longer-lived DNPs [16].

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2^2, \quad (9)$$

where

$$\begin{aligned} \tau_k &\in [1 \text{ ms}, 1000 \text{ s}] \\ \bar{\nu}_{d,k} &\in [0, 1]. \end{aligned}$$

MoSDeN uses the SciPy package's optimize module to solve this non-linear least squares problem [33].



Uncertainties are tracked using Monte Carlo sampling

The process for uncertainty tracking is as follows:

- 1 Normally sample $P_{n,i}$, τ_i , and $N_i(t)$ within their uncertainties for every DNP
- 2 Calculate the delayed neutron count rate ($\dot{n}_d$)
- 3 Calculate the DNP group parameters using non-linear least squares
- 4 Record the results and repeat

After all simulations finish:

- 1 Groups are defined based on half-lives (e.g. longest lived half-life and yield are group one)
- 2 The mean values and standard deviations are calculated



MoSDeN uses a newly-derived form of the group fit for MSRs

Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\frac{F(\tau)}{F_0} = \sum_{j=0}^J H(\tau - j \cdot \tau_{net}) - H(\tau - j \cdot \tau_{net} - \tau_{in}) \quad (10)$$

where

$F(\tau)$ = the fission rate history $[\frac{\#}{s}]$.

H = the Heaviside function $[-]$

$$\tau_{net} = \tau_{in} + \tau_{ex}$$

= the total time per circulation $[s]$

$$J = \left\lfloor \frac{T}{\tau_{net}} \right\rfloor$$

= the number of recirculations $[\#]$

F_0 = the magnitude of the constant fission rate $[\frac{\#}{s}]$.



The 0D scaled model is implemented in MoSDeN

MoSDeN uses simulation parameters and external data to generate improved DNP parameters.

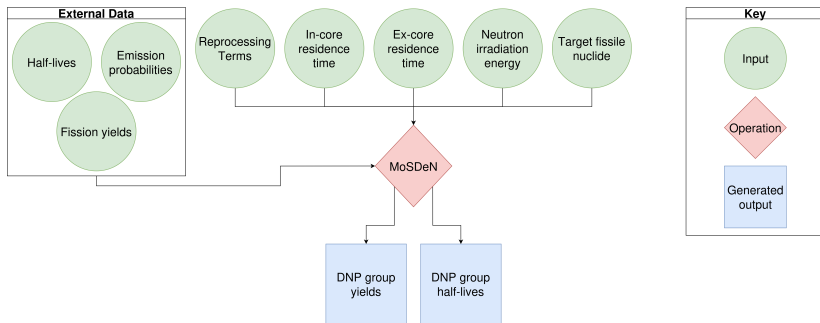


Figure 6: Flow diagram of 0D scaled model used to generate improved DNP group parameters.



The 0D scaled model is fast but uses many approximations

The equilibrium concentration of each DNP is calculated for the 0D scaled model as [13]:

$$N_i = \frac{f_{in} CFY_i}{\lambda_i}, \quad (11)$$

where

f_{in} = the fraction of time fuel-salt spends in the core [-]

CFY_i = the cumulative fission yield of nuclide i [-].

The limitations of the 0D scaled model include:

- A lack of absorption cross sections
- No depletion chain tracking for chemical removal
- Inability to model residence time effects



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The parameters for the 0D scaled model analyses

Table 1: Default parameters for 0D scaled model simulations.

Parameter	Value	Units
Fissile Nuclide	^{235}U	[-]
Incident Neutron Energy	0.0253	[eV]
Fission Yield Data	ENDF/B-VII.1 [6]	[-]
Half-life Data	IAEA [9]	[-]
Emission Probability Data	IAEA [9]	[-]
Final Decay Time	1200	[s]
Decay Time Spacing Method	Logarithmic	[-]
Decay Time Node Count	300	[-]
Sampled Simulation Count	1000	[-]
τ_{in}	9 [11]	[s]
τ_{ex}	16 [11]	[s]

The chemical removal rates for the 0D scaled model analyses

Table 2: MSBR Online Reprocessing Removal Rates from [23].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

The total delayed neutron yield is dominated by a few DNPs

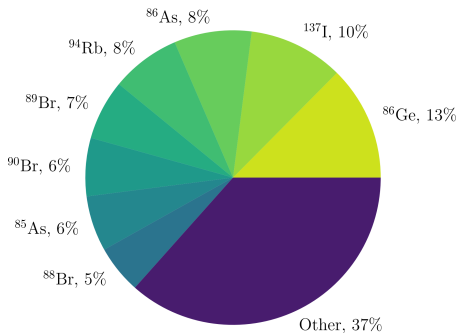


Figure 7: Relative delayed neutron yield from each DNP after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The delayed neutron count rate is dominated by a few DNPs

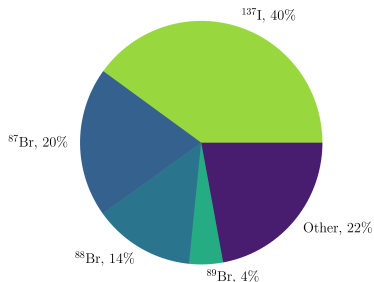


Figure 8: Relative delayed neutron counts from each DNP after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

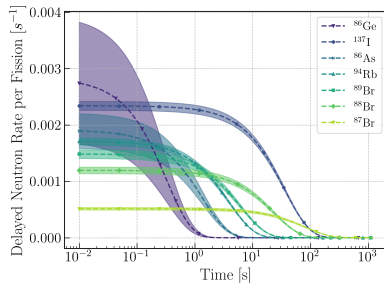


Figure 9: Relative delayed neutron count rates after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Chemical removal and simulation parameters affect the results

Sensitivity studies of the chemical removal show:

- Only the noble metal and gas reprocessing groups have a non-negligible impact on DNP group parameters
- Chemical removal in the 0D scaled model decreases the total delayed neutron yield by 12 *pcm*, a 0.57% difference

Sensitivity studies of the simulation parameters show:

- Logarithmic time node spacing improves the DNP group parameter fit compared to linear spacing with the same number of nodes
- Increasing the number of time nodes up to ≈ 800 improves the results
- Increasing the length of time for delayed neutron count data collection up to ≈ 40 minutes improves results

PCCs convey sensitivity of DNP group parameters to data

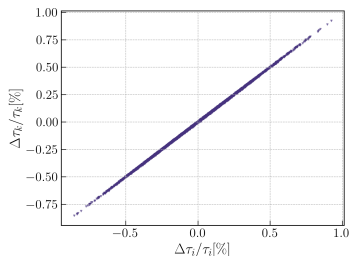


Figure 10: DNP group 1 half-life variation due to ^{87}Br half-life variation with a PCC of 1.00.

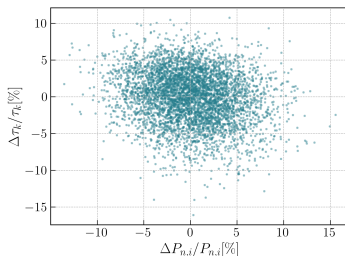


Figure 11: DNP group 3 half-life variation due to ^{87}Br emission probability variation with a PCC of -0.19.

The half-life of ^{87}Br strongly correlates to the group one half-life with a Pearson Correlation Coefficient (PCC) magnitude close to 1.0, while its emission probability does not strongly correlate to the group three half-life.

The PCC magnitudes show the most sensitive DNPs

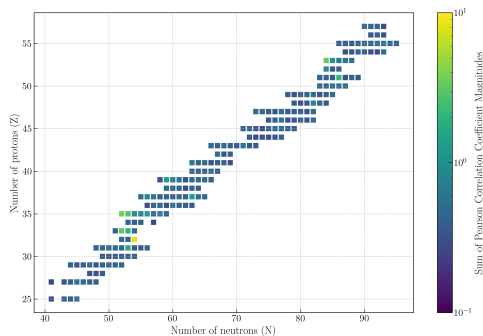


Figure 12: The summed PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Table 3: The largest summed PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nuclide	PCC_i
^{86}Ge	8.06
^{87}Br	3.82
^{137}I	3.52
^{85}As	3.14
^{88}Br	2.77
^{137}Sb	2.08
^{86}As	1.97
^{84}Ga	1.62
^{98}Y	1.42
^{100}Rb	1.22

The uncertainty-weighted PCC shows critical data needs

Uncertainty-weighted PCC U_i :

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_g |PCC_{i,v,g}|, \quad (12)$$

where

$U_{i,v}$ = the uncertainty-weighted sensitivity of
DNP value v for DNP i [-]

$\sigma_{i,v}$ = the uncertainty in the DNP value v
for DNP i [-]

$x_{i,v}$ = the DNP value v , representing the half-life,
emission probability, or concentration of DNP i [-]

$PCC_{i,v,g}$ = the PCC for DNP i and value v with
DNP group parameter g [-].

Table 4: The largest summed uncertainty-weighted PCC magnitudes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nuclide	U_i
^{86}Ge	1.71
^{124}Ag	0.41
^{120}Ag	0.36
^{114}Tc	0.36
^{147}Xe	0.34
^{104}Nb	0.29
^{111}Mo	0.27
^{122}Pd	0.27
^{98}Y	0.26
^{103}Zr	0.26

The DNP group parameters accurately represent the DNP data

The total delayed neutron yield ($\bar{\nu}_d$) from the DNP group parameters should align with the DNP data:

$$\sum_{k=1}^K \bar{\nu}_{d,k} \approx \sum_{i=1}^I P_{n,i} CFY_i = \bar{\nu}_d \quad (13)$$

The K groups give $\bar{\nu}_d = 2.24 \pm 0.13$ per 100 fissions, while the I DNPs give $\bar{\nu}_d = 2.25 \pm 0.15$.

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The average half-life ($\bar{\tau}$) from the DNP group parameters should align with the DNP data:

$$\ln(2) \sum_{k=1}^K \frac{\bar{\nu}_{d,k}}{\lambda_k} \approx \ln(2) \sum_{i=1}^I \frac{P_{n,i} CFY_i}{\lambda_i} = \bar{\tau} \quad (14)$$

The K groups give $\bar{\tau} = 6.4 \pm 0.3s$, while the I DNPs give $\bar{\tau} = 6.4 \pm 0.4s$.

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The K groups give $\bar{\tau} = 6.4 \pm 0.3s$, while the I DNPs give $\bar{\tau} = 6.4 \pm 0.4s$.

The total delayed neutron yield and average half-life agree between the model and the data.



The DNP group parameters do not align with the literature

The total delayed neutron yield ($\bar{\nu}_d$) from the DNP group parameters should align with the literature [21, 26, 6, 24, 20, 3, 7, 16, 8, 25, 29, 32, 34]:

The K groups give $\bar{\nu}_d = 2.24 \pm 0.13$ per 100 fissions, while the literature gives $1.48 \leq \bar{\nu}_d \leq 1.91$.

The average half-life ($\bar{\tau}$) from the DNP group parameters should align with the literature [24, 20, 3, 16, 8, 25, 29, 32, 34]:

The K groups give $\bar{\tau} = 6.4 \pm 0.3s$, while the literature gives $7.7 \leq \bar{\tau} \leq 9.5s$.

This indicates an issue with short-lived DNPs, likely from the selected nuclear data and the 0D scaled model assumptions.

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The objectives of this work are to:

-  Create a Python package (MoSDeN) for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems



The proposed work consists of three categories

Development of the 0D flow model:

- Provides new analyses and improves accuracy compared to the 0D scaled model
- Includes development of the MoSDeN Python package

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Validation of the improved DNP group parameters:

- Ensures the accuracy of the 0D flow model
- Enables comparison between the 0D flow and 0D scaled models

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Validation of the improved DNP group parameters:

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Analysis of safety, security, and safeguards using MSRE transients:

- Demonstrates the effects of the improved DNP group parameters
- Investigate how conventional DNP group parameters may predict differing behavior than the improved parameters

The 0D flow model will introduce new features and analyses

New features in the 0D flow model:

- Full depletion chain modeling in OpenMC
 - Improving chemical removal modeling
 - Enabling residence time analysis
 - Enabling non-equilibrium DNP concentrations
 - Improving DNP concentration accuracy
- Calculation of the improved DNP group spectra ($\chi_{d,k}$)

New analyses using the 0D flow model:

- Novel analysis for residual minimization using pulse and saturation irradiation data [4, 16]
- Comparison of various nuclear datasets [13, 21, 19]

The MSRE will be used for validation

Necessary components will be computed/generated:

- Improved DNP group parameters for the MSRE
- Multi-group cross sections using OpenMC for the MSRE
- Upper and lower plena added to an existing MSRE Moltres model

Quantities of interest will be computed and compared with experimental data for validation:

- Calculation of β_{eff} using the prompt method for stationary and flowing fuel salt using improved and conventional DNP group parameters [12]
- Simulate pump start-up and coast-down transients to compare reactivity responses to experimental data [22]

The MSRE will be used for validation

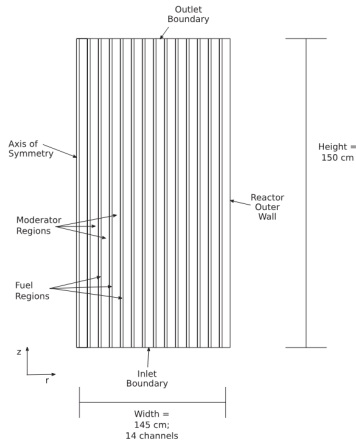


Figure 13: A sketch of the 2D MSRE Moltres model geometry (from [18]).

Safety-related transients may disrupt safe reactor operation

Safety-related transients that will be modeled are:

- Unprotected loss of heat sink (ULOHS)
- Unprotected loss of flow (ULOF)
- Unprotected transient overpower (UTOP)
- Fuel salt overcooling (FSOC)

Table 5: Implementation of transients using temperature, reactivity, and velocity changes

Transient	T	ρ	v
ULOHS	$T_{in} = T_{out}$	-	-
ULOF	-	-	$v \approx e^{-t}$
UTOP	-	$\rho_{max} = 0.1\%$	-
FSOC	$T_{in} = 0.9 T_{in,0}$	-	-

The safety transients will have different temperature and power responses between the improved and conventional DNP group parameters.

Sabotage that disrupts operation is a security concern

This framework assumes a hostile actor and an insider with the objective to disrupt operations [31].

Sabotage-initiated transients that will be modeled are:

- Combined UTOP and ULOHS
- Combined UTOP and ULOF
- Combined UTOP and FSOC

Table 6: Implementation of transients using temperature, reactivity, and velocity changes

Transient	T	ρ	v
ULOHS	$T_{in} = T_{out}$	-	-
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FSOC	$T_{in} = 0.9T_{in,0}$	-	-

The security transients will have different temperature and power responses between the improved and conventional DNP group parameters.

SNM diversion may be detected using reactor kinetics

Special nuclear material (SNM) diversion scenarios considered are [31, 14, 1]:

- One significant quantity (SQ) of ^{235}U (75 kg) after one year in the MSRE
 - Removing ^{235}U will decrease the delayed neutron fraction
- One SQ of Pu (8 kg) after one year in the MSRE
 - Removing Pu will increase the delayed neutron fraction

The DNP group parameters will then be used in a negative reactivity step insertion, and the resulting power responses will be compared [35].

Each scenario will have four sets of DNP group parameters to compare:

- Conventional parameters with/without diversion
- Improved parameters with/without diversion

The change in fuel composition and the MSR phenomena affect the ability to detect SNM diversion.

Summary

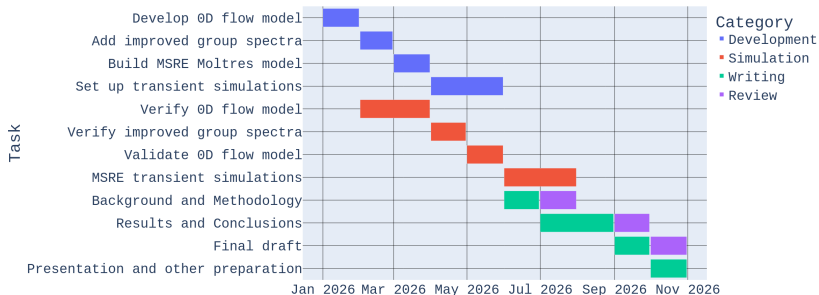


Figure 14: A Gantt chart of the proposed work.

References I

- [1] Charles G. Bathke, Bartley B. Ebbinghaus, Brian A. Collins, Brad W. Sleaford, Kevin R. Hase, Martin Robel, Richard K. Wallace, Keith S. Bradley, John R. Ireland, Gordon D. Jarvinen, M. W. Johnson, Andrew W. Prichard, and Brian W. Smith.

The attractiveness of materials in advanced nuclear fuel cycles for various proliferation and theft scenarios.

Nuclear Technology, 179(1):5–30, 2012.

- [2] Benjamin Betzler.

Liquid-fueled molten salt reactor depletion modeling.

Technical Report OSTI 1809981, Oak Ridge National Laboratory / U.S. Department of Energy, Oak Ridge, TN, USA, 2021.

- [3] MC Brady and TR England.

Delayed neutron data and group parameters for 43 fissioning systems.

Nuclear Science and Engineering, 103(2):129–149, 1989.

- [4] Michaele Clarice Brady.

Evaluation and application of delayed neutron precursor data.

PhD thesis, Texas A&M University, 1988.

References II

- [5] GS Brunson, EN Pettitt, and RD McCurdy.
Measurement of Delayed Neutron Yields in Plutonium, Uranium-233, Uranium-238, and Thorium Relative to Yield in Uranium-235.
Technical Report 5480, Argonne National Lab., Lemont, Ill., 1955.
- [6] Mark B Chadwick, Michal Herman, P Obložinský, Michael E Dunn, Yaron Danon, AC Kahler, Donald L Smith, Boris Pritychenko, Goran Arbanas, R Arcilla, et al.
ENDF/B-VII. 1 nuclear data for science and technology: cross sections, covariances, fission product yields and decay data.
Nuclear data sheets, 112(12):2887–2996, 2011.
- [7] MB Chadwick, P Obložinský, Michal Herman, NM Greene, RD McKnight, DL Smith, PG Young, RE MacFarlane, GM Hale, SC Frankle, et al.
ENDF/B-VII. 0: next generation evaluated nuclear data library for nuclear science and technology.
Nuclear data sheets, 107(12):2931–3060, 2006.

References III

- [8] William S Charlton, Theodore A Parish, Subramanian Raman, Nubuo Shinohara, and Masaki Andoh.
Delayed Neutron Emission Measurements from Fast Fission of U235 and Np237.
Technical report, Texas A & M Univ., College Station, TX (United States); Oak Ridge National Laboratory, 1996.
- [9] P. Dimitriou, I. Dillmann, B. Singh, V. Piksaikin, K.P. Rykaczewski, J.L. Tain, A. Algora, K. Banerjee, I.N. Borzov, D. Cano-Ott, S. Chiba, M. Fallot, D. Foligno, R. Grzywacz, X. Huang, T. Marketin, F. Minato, G. Mukherjee, B.C. Rasco, A. Sonzogni, M. Verpelli, A. Egorov, M. Estienne, L. Giot, D. Gremyachkin, M. Madurga, E.A. McCutchan, E. Mendoza, K.V. Mitrofanov, M. Narbonne, P. Romojaro, A. Sanchez-Caballero, and N.D. Scielzo.
Development of a reference database for beta-delayed neutron emission.
Nuclear Data Sheets, 173:144–238, 2021.
Special Issue on Nuclear Reaction Data.
- [10] JR Engel, PN Haubenreich, and A Houtzeel.
Spray, mist, bubbles, and foam in the molten salt reactor experiment.
Technical Report ORNL-TM-3027, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States), 1970.

References IV

- [11] S Hanauer.
Neutron-fluctuation measurements.
In Neutron Noise, Waves and Pulse Propagation: Proceedings, volume 9, page 463. US Atomic Energy Commission, Division of Technical Information, 1967.
- [12] Paul N Haubenreich and JR Engel.
Experience with the molten-salt reactor experiment.
Nuclear Applications and technology, 8(2):118–136, 1970.
- [13] TD Huynh and C Jouanne.
Calculation of delayed neutron yields for various libraries.
Nuclear Data Sheets, 118:476–479, 2014.
- [14] International Atomic Energy Agency.
IAEA Safeguards Glossary.
International Atomic Energy Agency, Vienna, 2022.
- [15] George Robert Keepin.
Delayed Neutrons: A Review as of October 1955.
Technical Report LA1970, Los Alamos National Laboratory, Los Alamos, NM, USA, 1970.

References V

- [16] George Robert Keepin, TF Wimett, and RK Zeigler.
Delayed neutrons from fissionable isotopes of uranium, plutonium and thorium.
Journal of Nuclear Energy (1954), 6(1-2):IN2–21, 1957.
- [17] Thomas W Kerlin and Belle R Upadhyaya.
Dynamics and control of nuclear reactors.
Academic press, 2019.
- [18] Alexander Lindsay, Gavin Ridley, Andrei Rykhlevskii, and Kathryn Huff.
Introduction to Moltres: An application for simulation of Molten Salt Reactors.
Annals of Nuclear Energy, 114:530–540, April 2018.
- [19] L Mathieu, Olivier Serot, T Materna, A Bail, U Köster, H Faust, O Litaize, E Dupont, C Jouanne, A Letourneau, et al.
New neutron long-counter for delayed neutron investigations with the lohengrin fission fragment separator.
Journal of Instrumentation, 7(08):P08029, 2012.

References VI

- [20] RW Mills, MF James, and DR Weaver.

Study of the delayed neutron yield and its time dependence by the summation method, and the sensitivity of the yield to parameters of the independent yield model and the decay data.

In *Nuclear Data for Science and Technology: Proceedings of an International Conference, held at the Forschungszentrum Jülich, Fed. Rep. of Germany, 13–17 May 1991*, pages 86–88. Springer, 1992.

- [21] NEA Data Bank.

The JEF-2.2 nuclear data library. JEFF report 17, 2000.

- [22] BE Prince, SJ Ball, JR Engel, PN Haubenreich, and TW Kerlin.

Zero-power physics experiments on the molten-salt reactor experiment.

Technical Report ORNL-4233, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States), 1968.

- [23] R.C. Robertson.

CONCEPTUAL DESIGN STUDY OF A SINGLE-FLUID MOLTEN-SALT BREEDER REACTOR.

Technical Report ORNL-4541, ORNL, January 1971.

References VII

- [24] P F Rose.
ENDF-201: ENDF/B-VI summary documentation.
Technical Report BNL-NCS-17541, Brookhaven National Lab., Upton, NY (United States), 10 1991.
- [25] Habeeb H Saleh, Theodore A Parish, S Raman, and Nobuo Shinohara.
Measurements of delayed neutron decay constants and fission yields from ^{235}U , ^{237}Np , ^{241}Am , and ^{243}Am .
Nuclear science and engineering, 125(1):51–60, 1997.
- [26] A Santamarina, Dea Bernard, P Blaise, M Coste, A Courcelle, TD Huynh, C Jouanne, P Leconte, O Litaize, S Mengelle, et al.
The JEFF-3.1.1 nuclear data library.
JEFF report, 22(10.2):2, 2009.
- [27] D Saphier, D Ilberg, S Shalev, and S Yiftah.
Evaluated delayed neutron spectra and their importance in reactor calculations.
Nuclear Science and Engineering, 62(4):660–694, 1977.

References VIII

- [28] Seifert, Luke, Shahbazi, Shayan, Richards, Scott, Munk, Madicken, and Huff, Kathryn.
A Computationally Tractable Model for Spatially Resolved Nuclide Concentrations in Flowing Fuel Molten Salt Reactors.
In Proceedings of The International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025), pages 92–101, Denver, Colorado, April 2025. American Nuclear Society.
- [29] S Synetos and JG Williams.
Delayed neutron yield and decay constants for thermal neutron-induced fission of ^{235}U .
Nucl. Energy (Inst. Nucl. Eng.), 22(4), 1983.
- [30] M Tano, S A Walker, and A Abou-Jaoude.
A Coupled Griffin and Pronghorn Method for Spatially Resolved Depletion Analysis of Molten Salt Reactors.
In American Nuclear Society Mathematics & Computation, 2023.
- [31] The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group.
Evaluation Methodology for Proliferation Resistance and Physical Protection of Generation IV Nuclear Energy Systems.
Technical Report GIF/PRPPWG/2011/003, Generation IV International Forum, 2011.

References IX

- [32] RJ Tuttle.
Delayed-neutron data for reactor-physics analysis.
Nuclear Science and Engineering, 56(1):37–71, 1975.
- [33] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors.
SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python.
Nature Methods, 17:261–272, 2020.
- [34] RW Waldo, RA Karam, and RA Meyer.
Delayed neutron yields: Time dependent measurements and a predictive model.
Physical Review C, 23(3):1113, 1981.
- [35] Alexander M Wheeler, Ondřej Chvála, and Steven Skutnik.
Signatures of plutonium diversion in molten salt reactor dynamics.
Annals of Nuclear Energy, 160:108370, 2021.

MoSDeN uses a newly-derived form of the group fit for MSRs



Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\begin{aligned}
 \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{\nu}_{d,k}} = & \overbrace{e^{-\lambda_k t} \sum_{j=0}^{J_{in}} \left(e^{-\lambda_k (T-j \cdot \tau_{net} - \tau_{in})} - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}^{\text{Partial ex-core}} \\
 & + \underbrace{e^{-\lambda_k t} \sum_{j=J_{in}+1}^J \left(1 - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}_{\text{Partial in-core}}
 \end{aligned} \tag{15}$$

where

T = the total irradiation time

$$J_{in} = \left\lfloor \frac{T - \tau_{in}}{\tau_{net}} \right\rfloor$$

= the number of recirculated in-core irradiations [#]

Long cycle time elements have a negligible impact



Group half-lives:

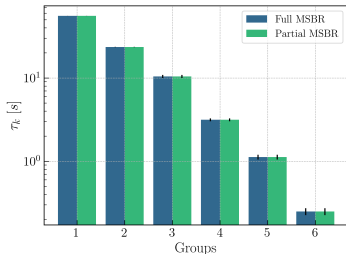


Figure 15: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

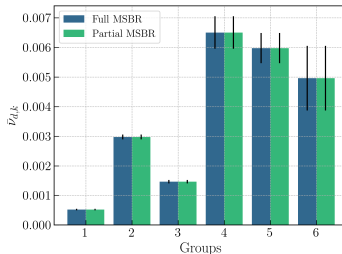


Figure 16: $\bar{\nu}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The differences induced by exclusion of removal rates with cycle times longer than 20 seconds are minimal and well within uncertainty bounds.

Removal rates have a small impact in the 0D scaled model



Group half-lives:

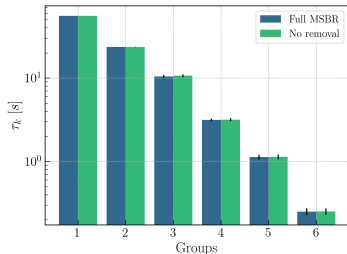


Figure 17: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

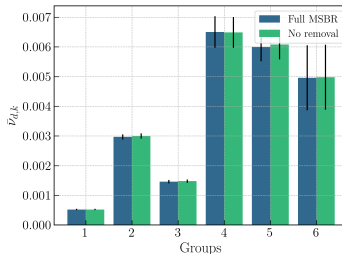


Figure 18: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Neglecting chemical removal rates generally leads to larger yields, though all differences are within uncertainty bounds.

Logarithmic spacing of time nodes give more accurate results



Group half-lives:

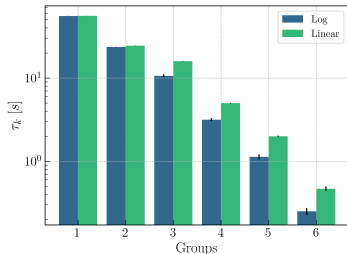


Figure 19: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

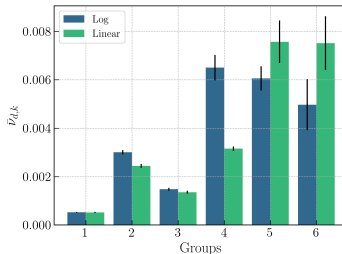


Figure 20: $\bar{\nu}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution.

More time nodes give slightly more accurate results



Group half-lives:

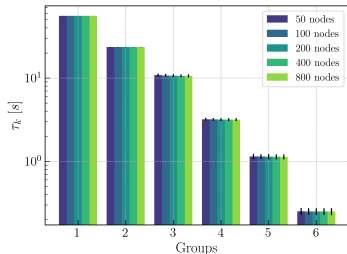


Figure 21: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

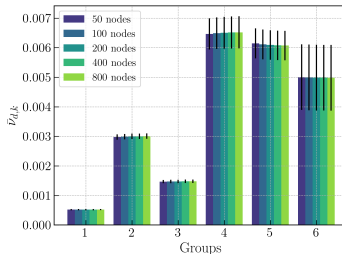


Figure 22: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Increasing the number of time nodes increases the accuracy of the DNP group parameters, with differences within uncertainty bounds.

Longer measurement times give more accurate results



Group half-lives:

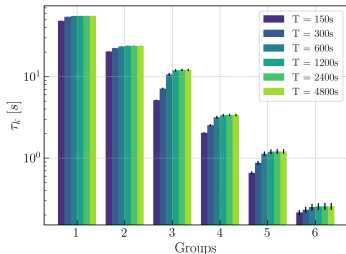


Figure 23: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

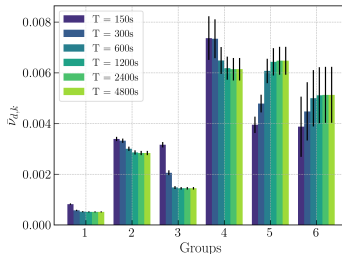


Figure 24: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Longer post-irradiation time results in longer-lived groups with smaller yields in groups one through four and larger yields in groups five and six.

The density of time nodes does not drive the group parameters



Group half-lives:

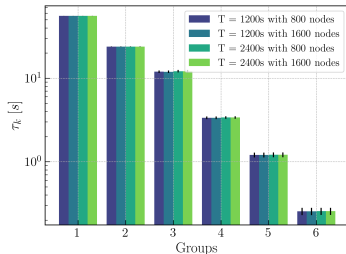


Figure 25: τ_k for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Group yields:

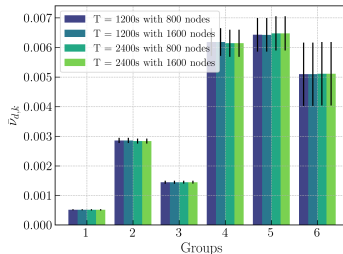


Figure 26: $\bar{v}_{d,k}$ for each DNP group k after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The number of nodes has a small impact on the DNP group parameters, whereas the total decay time causes larger differences.

Safety-related transients may disrupt safe reactor operation



Safety-related transients that will be modeled are:

- ULOHS
 - Loss of cooling in the reactor, potentially due to a heat exchanger failure
 - Modeled in Moltres by setting inlet temperature equal to outlet temperature
- ULOF
 - Flow rate decreases to zero due to a pump failure
 - Modeled in Moltres by changing flow field over time to match MSRE pump shut-down
- UTOP
 - Reactivity insertion event, which could be caused by poison removal, fissile addition, or temperature change
 - Modeled in Moltres by having a step-decrease in the removal cross section, representing control rod withdrawal
- FSOC
 - Fuel salt temperature decreases, potentially caused by increased heat removal rate
 - Modeled in Moltres by reducing inlet temperature by 25%

Sabotage may be used as a diversion

This framework assumes a hostile actor and an insider with the objective to disrupt operations [31].

Sabotage-initiated transients that will be modeled are:

- All safety-related transients as single events
- Combined UTOP and ULOHS
 - This should cause a temperature increase that propagates due to lack of heat sink; reactivity should then decrease due to temperature increase
- Combined UTOP and ULOF
 - This should cause a temperature increase with a peak maximum temperature due to the stationary fuel salt; reactivity should then decrease due to temperature increase
- Combined UTOP and FSOC
 - This should cause a temperature increase that is counteracted by the overcooled inlet fuel salt, leading to the largest reactivity change; temperature is not expected to change dramatically